Visualization of Formula One Career Results

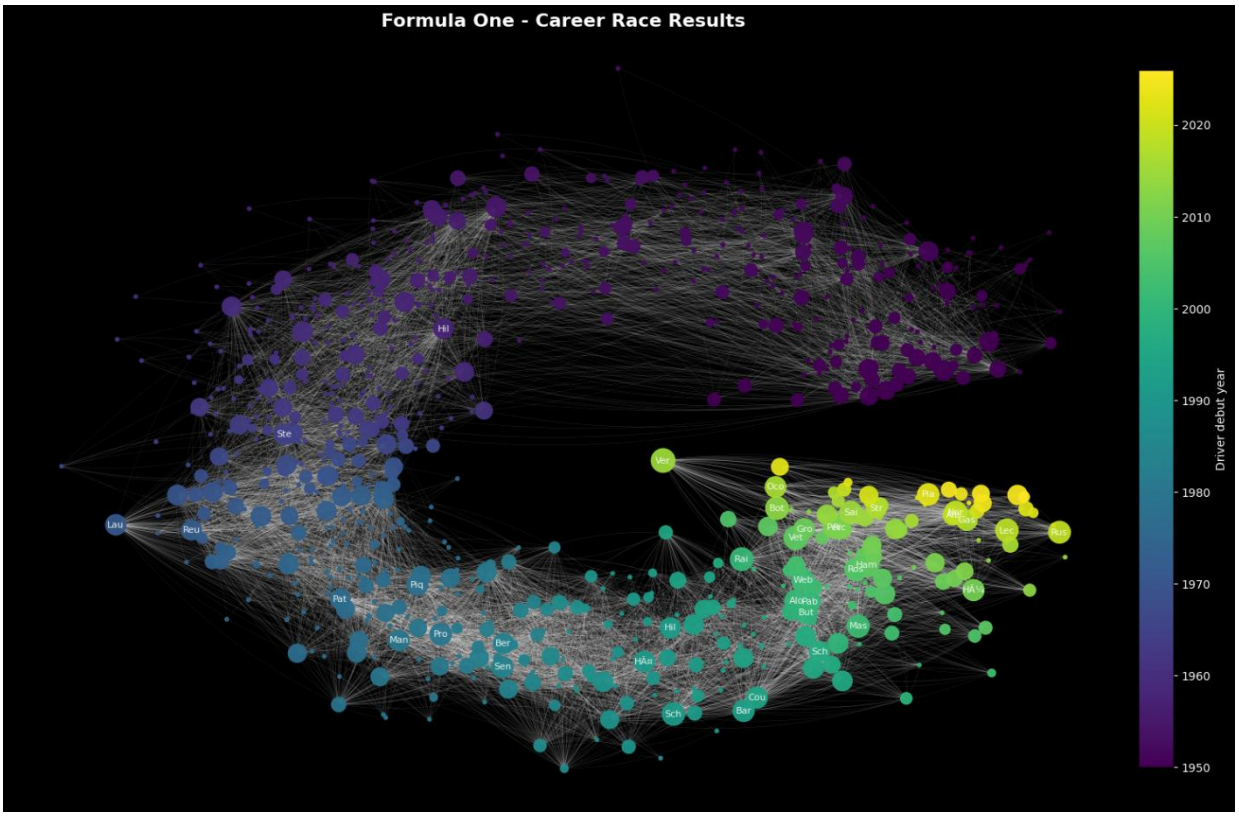

Fig. 1: Formula One Career Results

Introduction: Formula One is considered the highest class of international in open-wheel single-seat racing. Having just celebrated the 75th anniversary of the sport, the sport has had consistent formatting of 20 drivers and 10 teams (2 drivers per team) with a varying number of races per season. However, the 2026 season introduced an 11th team, making it a total of 22 drivers per season. Based on the finishing position of each race, the top 10 drivers are awarded points.

Dataset: The Formula One Race Data[1] used for this project was sourced through Kaggle. The dataset contains 14 different tables with data ranging from seasons beginning in 1950 and through 2025. With the use of Microsoft Access, I combined 5 of the available tables (Results, Drivers, Race Status, Races, Lap Times) to create a consolidated table[2] of 27,304 records. Instances of a race status outside of "Finished" were not removed because a status of retired or DNF still contributes to race result and position. The inspiration for the structure of the graph comes from the YouTube video " I Made a Network of Every Home Run in MLB History" by adumb (link)

Visualization and Analysis

In figure 1, Nodes represent driver initial debut, with earliest (1950) = purple, latest (2025) = yellow.

- The edges depict race results of each driver; the connection is linked to the driver that finished in the position directly behind them (ex. Driver 1 finished 1st -> Driver 2 finished 2nd).
- Nodes are clustered based on common links. If they finish next to each other often, they will be closer together.

- Node size is determined by a driver's total career points; larger nodes are more successful.
- There is a distinct gap between purple nodes and yellow nodes due to the average career length of a Driver consisting of 3-5 years, making overlap impossible

Figure 2 shows the top 40 drivers with the longest careers

- Calculated based on debut season and final season
- As we the year increase, career length has increased
- Drivers that debuted in 2000s have had a higher average of career length

Figure 3 shows each of the top 40 drivers and their strongest connection.

- Strength is determined by edge weight (number of connections)
- Earlier nodes (purple) we see much longer connections, indicating longer career length
- Consistent performance increases as the year goes up; links are thicker

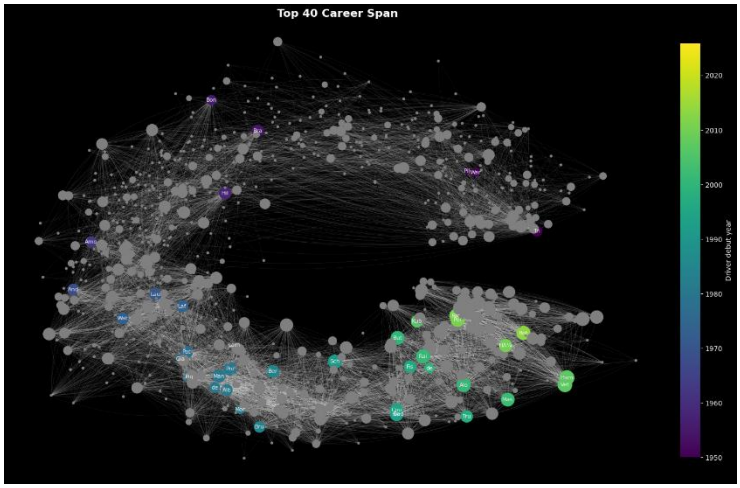

Fig. 2: Top 40 Drivers with longest career span

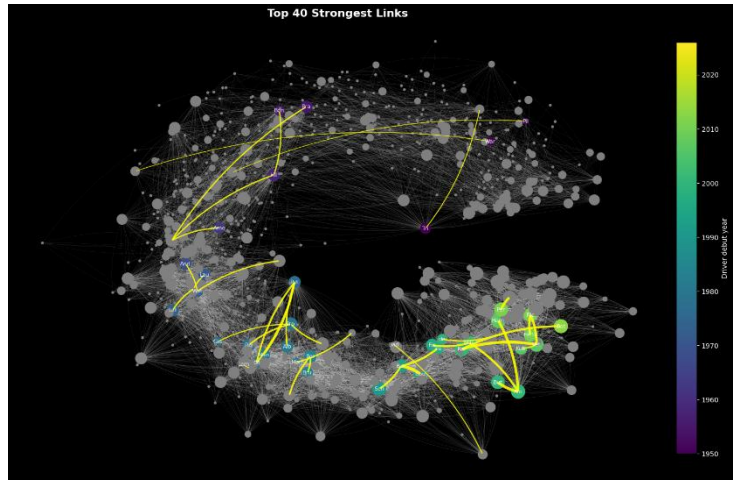

Fig. 3: Top 40 Drivers strongest links

In Fig 4. The top 40 drivers' maximum points earned in one season vs. career span

- Analyze if there is corresponding relationship between career length and peak performance.
- There is an average of about 12-year career spans for the top 40 drivers
- most do not reach season performance peaks of above 100 points.
- Higher max season points are drivers that have debuted within the last ~20 years

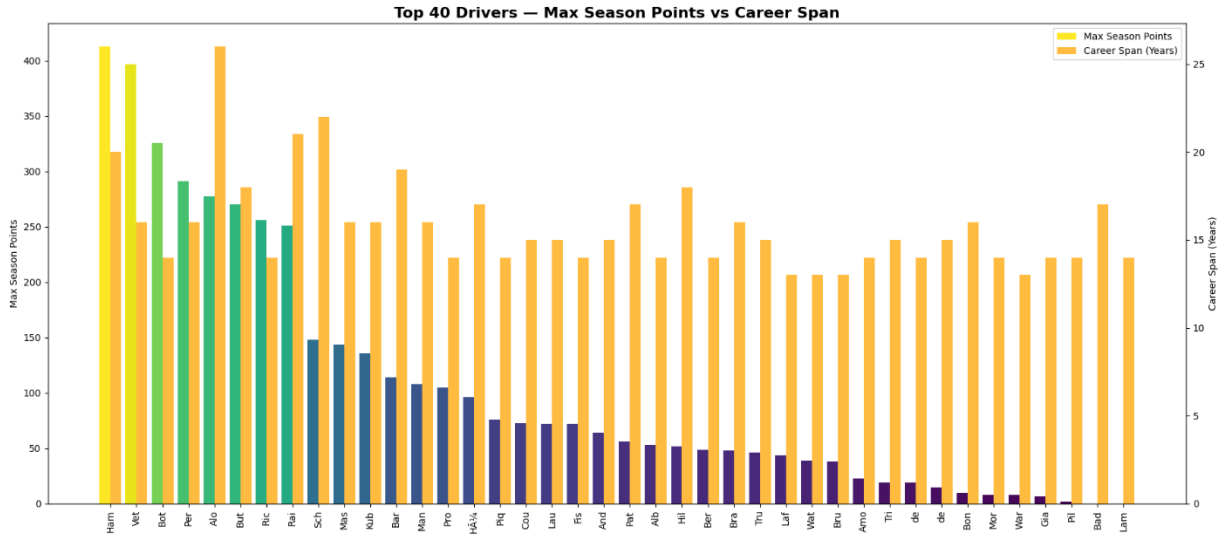

Fig. 4: Lifetime Points of Top 40 Longest Career Drivers

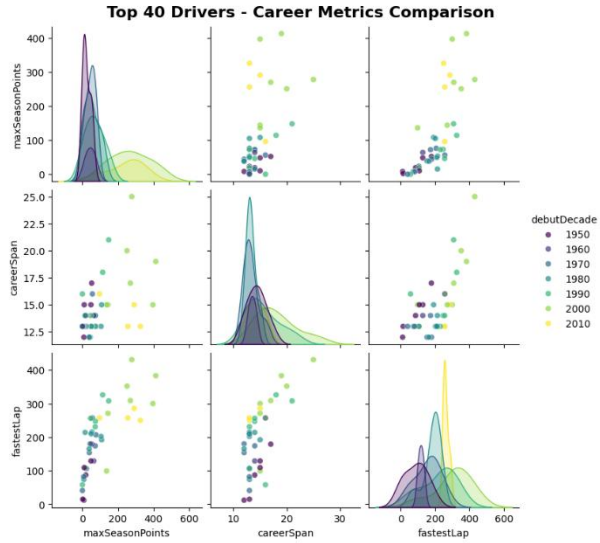

Fig. 5: top 40 Career Metric Comparison with respect to debut decade

Figure 5 depicts career metric comparison between the top 40 longest career drivers.

- Increase in fastest lap leads to increase in max season points
- Career span generally shows high max season points, though drivers debuting in 2000 or 2010 have significantly higher max season points regardless of career span
- Drivers debuted in decades 1950 through 1970, generally perform worse than those debuted after
- Generally, the average career span for decade 2000 is 12 years, but there is an outlier (ALO) that drags the distribution.

Conclusion: As we approach the year 2025, we see an increase in node size indication that general performance of drivers is increasing. This is likely an indication of increased talent but also an increase in machinery capabilities. Additionally, strongest links decrease in length over time, this is an indicator of consistent finishing, showing drivers have more direct rivals, despite drivers with earlier debut having a higher average of career length. Career length is not an indicator of peak performance, as we observed the top 40 career length drivers and found that most had not reach a peak of100 points in one season. This is a sport that is heavily dependent on technology so it makes sense that there will be an increase in performance as years increase following technological advancements in sport.

Importance: There is a lot of debate within the sport about the exclusivity of the sport, which leads to shorter career spans. Many fans get upset when their favorite driver gets replaced and feel it is unfair. But to keep the reputation of "the pinnacle of motorsport" it is important to prioritize performance over "fairness" and this helps to understand that historically a driver can have a successful career both by maximizing points per season and by being a consistent mid-tier driver.

References

[1] https://www.kaggle.com/datasets/jtrotman/formula-1-race-data?select=circuits.csv
[2] Consolidated data "Results_4.xlxs"
[3] https://youtu.be/LtJxJqAqt0Q?si=Zws-8BVfBcJ-Nqno
[4] https://github.com/Hansen411/Final-Project/tree/main